

ENCI335

Structural Analysis and Systems I

Tutorial 5 Solution

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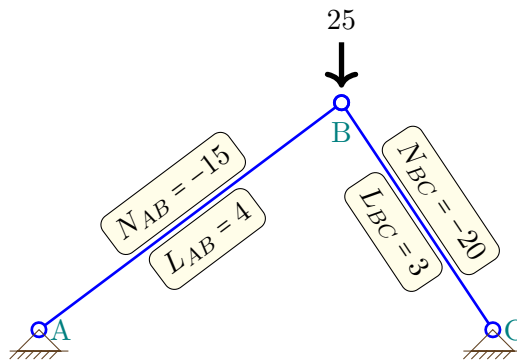
Objectives

This tutorial provides an opportunity for you to practice the following:

- Energy Principle
- Betti's Theorem

Question 1 Energy Principle

Consider the following truss structure with all members having the same axial rigidity EA . Due to the vertical load of 25 acting at B, the compressive axial forces in members AB and BC are 15 and 20, respectively. Use the principle of conservation of energy $W = U$ to determine the vertical deflection at B of the truss structure in terms of EA .



Solution:

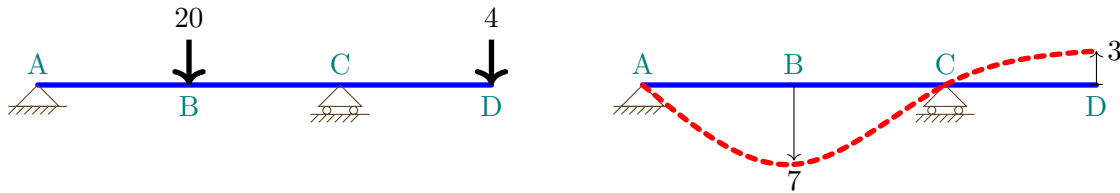
$$W = \frac{1}{2} \cdot (-25) \cdot v_B \quad (1)$$

$$U = \sum \frac{N^2 L}{2EA} = \sum \frac{1}{2EA} (15^2 \cdot 4 + 20^2 \cdot 3) = \frac{1050}{EA} \quad (2)$$

$$W = U \implies v_B = \frac{-84}{EA} (\downarrow). \quad (3)$$

Question 2 Energy Principle 2

An overhanging beam is subjected to two point loads at B and D, as shown below. Under these loads, the beam deflects as shown. Calculate the total work done by these loads during this deformation.



Solution:

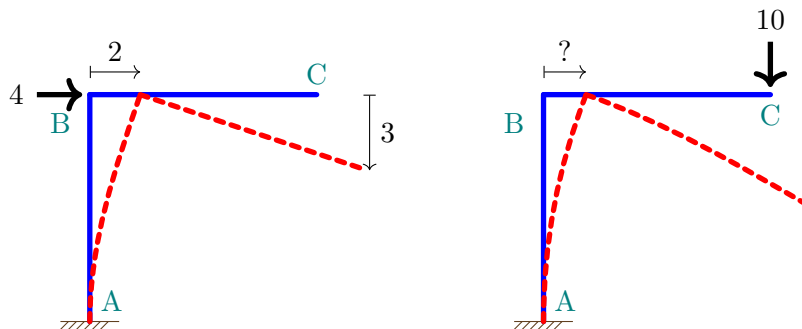
$$W = \frac{1}{2} \cdot P_B \cdot v_B + \frac{1}{2} \cdot P_D \cdot v_D \quad (4)$$

$$= \frac{1}{2} \cdot (-20) \cdot (-7) + \frac{1}{2} \cdot (-4) \cdot 3 \quad (5)$$

$$= 64 \quad (6)$$

Question 3 Betti's Theorem

Consider the L-frame shown below. When a horizontal point load of 4 units is applied at B, the horizontal displacement at B is 2 units, while the vertical deflection at C is 3 units. Using Betti's theorem, determine the horizontal displacement at B when a vertical point load of 2 units is applied at C.



Solution:

According to Betti's law,

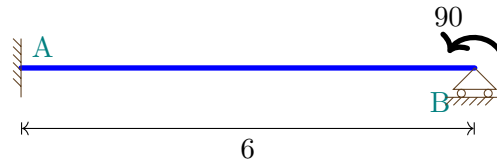
$$P_B \cdot v_{BC} = P_C \cdot v_{CB} \quad (7)$$

$$\implies 4 \cdot v_{BC} = (-10) \cdot (-3) \quad (8)$$

$$\implies v_{BC} = 7.5 \quad (9)$$

Question 4 Betti's Theorem and Energy Principle

Consider the propped cantilever beam shown below.

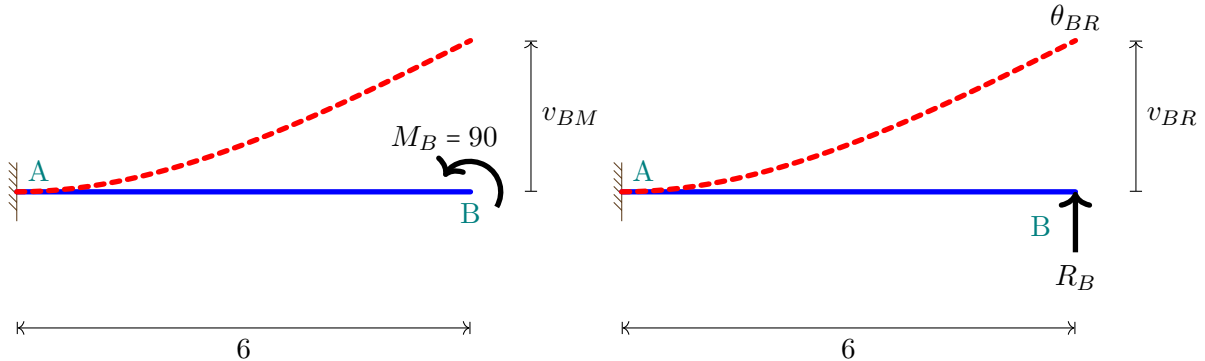


Using only the information provided in the table below, determine all reaction forces and use the energy principle to determine the rotation at B.

Beam and Loading	Elastic Curve	Maximum Deflection	Slope at End
1		$-\frac{PL^3}{3EI}$	$-\frac{PL^2}{2EI}$

Solution:

The support reaction at B can be treated as an unknown redundant load, denoted as R_B . The following two loading conditions and their assumed deflected shapes are then considered.



The compatibility condition requires zero deflection at B, i.e.,

$$v_{BM} + v_{BR} = 0 \quad (10)$$

where the second subscript, M or R , indicates the load case: M for the point moment M_B and R for the redundant force R_B .

According to Betti's law:

$$R_B v_{BM} = M_B \theta_{BR} = 90 \theta_{BR} \quad (11)$$

$$\implies v_{BM} = \frac{90}{R_B} \theta_{BR} \quad (12)$$

Substitute the above into the compatibility condition gives:

$$\frac{90}{R_B} \theta_{BR} + v_{BR} = 0 \quad (13)$$

which, after rearrangement, yields

$$R_B = -\frac{90\theta_{BR}}{v_{BR}} \quad (14)$$

According to the table provided, $\theta_{BR}/v_{BR} = 3/(2L)$. Therefore,

$$R_B = -22.5(\downarrow) \quad (15)$$

To find the rotation at B, we use the energy principle, where $W = \frac{1}{2}M_B\theta_B$, and $U = \int_0^L \frac{M(x)^2}{2EI} dx$. With $R_B = -22.5$, the bending moment distribution is established as:

$$M(x) = 90 - 22.5x \quad (16)$$

Therefore

$$W = U \quad (17)$$

$$\implies \frac{1}{2}M_B\theta_B = \int_0^L \frac{M(x)^2}{2EI} dx \quad (18)$$

$$\implies \theta_B = \frac{2}{M_B} \cdot \int_0^L \frac{(90 - 22.5x)^2}{2EI} dx = \frac{135}{EI} \quad (19)$$